

Sudhanshu Singh

Software Development Engineer in Test (SDET)

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PROFESSIONAL SUMMARY

SDET with 4+ years of experience designing and delivering scalable automation frameworks for web, mobile, and hardware-integrated applications. Proficient in WebdriverIO, Playwright, Appium, and TestCafe using TypeScript, JavaScript, and Python. Known for solving complex testing challenges — including developing a Windows-based virtual USB/HID device simulation using the UDE framework to enable hardware-independent telephony automation. Experienced in CI/CD integration via Azure DevOps and cross-platform mobile testing through BrowserStack, with a consistent record of accomplishment of improving quality and accelerating release cycles in Agile environments.

TECHNICAL SKILLS

Testing Frameworks	: WebdriverIO, Playwright, Appium, TestCafe, Cucumber (Gherkin), pytest
Automation Tools	: REST Assured, Postman, MSTest, NUnit, Axios
Programming Languages	: TypeScript, JavaScript, Python, Java, C#
CI/CD & Cloud	: Azure DevOps, GitHub, BrowserStack (Web & Mobile)
IDEs & Editors	: Visual Studio, VS Code, Cursor, IntelliJ IDEA, PyCharm, PowerShell
Testing Types	: UI Automation, Mobile (Android & iOS), API, Cross-Browser & Regression
AI Experties	: Prompt Engineering, GitHub Copilot, Claude, OpenAI

PROFESSIONAL EXPERIENCE

Software Development Engineer in Test (SDET) | HCLTech *Mar 2023 – Present · Lucknow, India*

- Designed and maintained scalable automation frameworks using WebdriverIO, Playwright and TestCafe, improving regression stability across sprint releases and reducing manual validation effort by ~60%.
- Implemented mobile automation testing for Android and iOS applications using Appium across BrowserStack cloud environments, covering cross-device and cross-browser validation at scale.
- Automated external HID device interaction workflows validating telephony call-control scenarios (incoming calls, mute synchronization, hold/resume, volume control) for a unified communication desktop application.
- Integrated automation suites with Azure DevOps CI/CD pipelines, enabling continuous regression execution and reducing release validation time by ~40%.
- Developed reusable automation utilities and modular test components, reducing script duplication by ~35%, and tracked defects via Azure DevOps Boards to ensure timely resolution and accurate sprintlevel quality reporting.

Engineering Intern | HCLTech *Mar 2022 – Feb 2023 · Lucknow, India*

- Developed responsive web interfaces using HTML, CSS, and JavaScript, and gained hands-on exposure to ReactJS, Angular, and backend development using C# and .NET within MVC architecture.
- Assisted the QA team in manual testing activities including test case writing, bug reporting, and regression verification, building foundational understanding of SDLC and release processes.
- Promoted to full-time SDET at HCLTech upon internship completion, directly applying QA and development foundations to build and scale automation frameworks from the ground up.

KEY ENGINEERING PROJECTS

Virtual Device Simulation for Automation Testing

- Built a Windows-based virtual USB composite device (Audio + HID) using the UDE framework, requiring one-time driver installation to enable hardware-independent telephony automation locally.
- Developed KMDF-based virtual HID architecture and a Node.js integration layer using IOCTL interface, allowing automation frameworks to trigger HID signals programmatically via REST APIs.
- Eliminated dependency on physical headsets for validating call-control scenarios (incoming calls, mute sync, hold/resume, volume control) without cloud or BrowserStack support.

Automation Framework for Hardware-Integrated Applications (WebdriverIO)

- Architected a WebdriverIO + TypeScript framework using Page Object Model with retry logic, parallel execution, and environment-specific configuration for local and CI contexts.
- Integrated with Azure DevOps pipelines for nightly regression runs and built reusable modular components for consistent cross-environment test execution.

Web Automation Testing (Playwright)

- Built a scalable Playwright automation framework supporting cross-browser execution (Chromium, Firefox, WebKit) with parallel test execution and CI integration.
- Automated complex workflows including authenticated flows, dynamic routing and shadow DOM interactions using reusable fixtures and robust locator strategies, improving regression stability.

Mobile Automation Testing (Appium & BrowserStack)

- Developed an Appium test suite for Android and iOS covering gestures, deep links, and session handling, with a shared architecture managing platform-specific locator differences.
- Configured BrowserStack with parameterized device and OS version matrices, enabling scalable crossdevice validation without a physical device lab.

AWARDS & RECOGNITION

HCL ERS Spot Award · HCLTech — Recognized for consistently dedicated efforts.

HCL ERS Champion · HCLTech — Recognized for providing the maximum number of ideas resulting in significant value and cost savings for the project.

Marathon Mind Award · HCLTech ERS — Recognized for proposing ideas driving innovation with Virtual Device Simulation and GenAI.

EDUCATION

Bachelor of Computer Applications (BCA)

Amity University, Noida, India

Intermediate — PCM

DAV Centenary Public School, Bhawnathpur, India